

Primera Entrega - Proyecto

Santiago Cadena, Camilo Chacón, Felipe Ortiz

Departamento de Ingeniería Eléctrica y Electrónica
Electrónica Digital II

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1 Funcionamiento de protocolo VGA

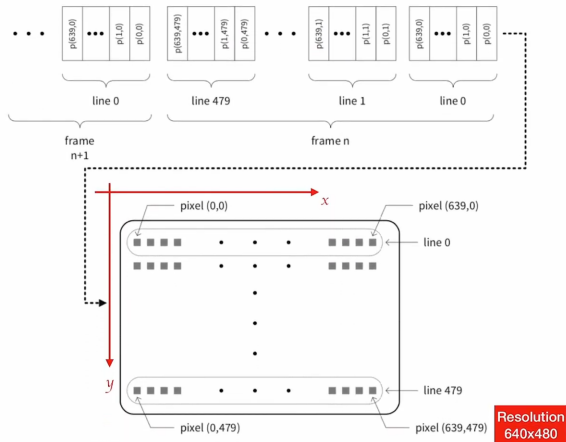
- Pixeles y frames
- Stream interface
- FIFO buffer
- CTR Monitor
- Horizontal synchronization
- Vertical synchronization
- Pixel clock rate
- VGA
- Tiles
- Matriz

2 SoC - Space Invaders

3 Mapa de Memoria



VGA (Video Graphics Array)

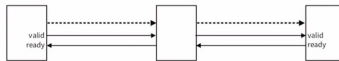


VGA (Video Graphics Array)

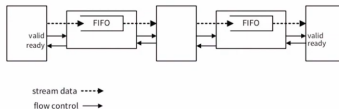
- Centralized flow control



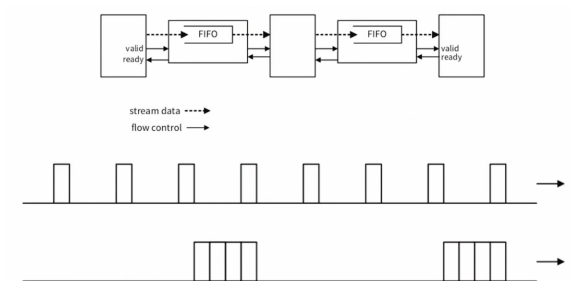
- Distributed flow control with back pressure



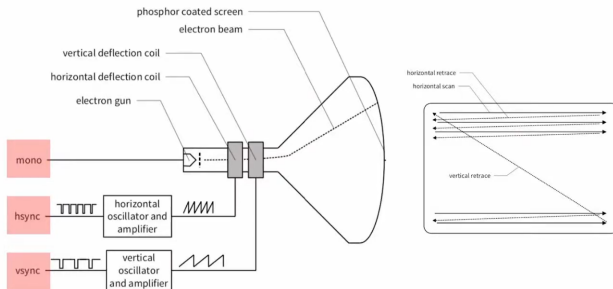
- Distributed flow control with a FIFO buffer



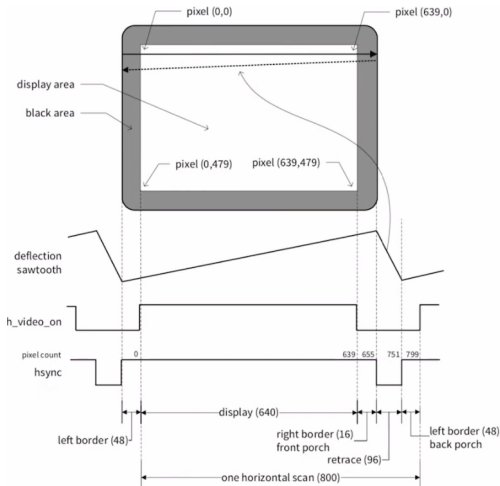
VGA (Video Graphics Array)



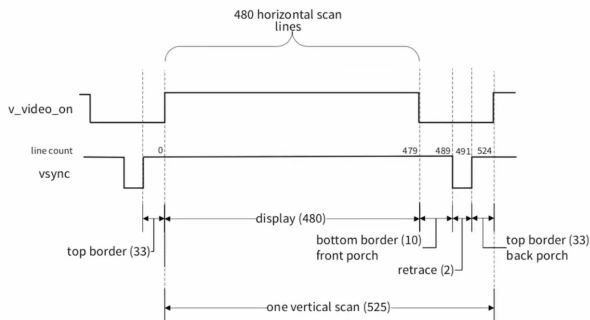
VGA (Video Graphics Array)



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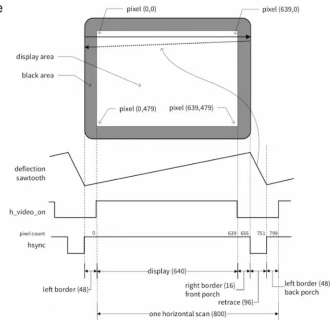


VGA (Video Graphics Array)

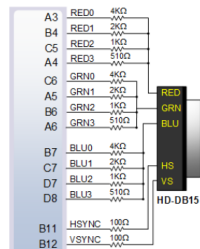
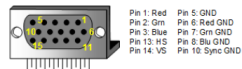
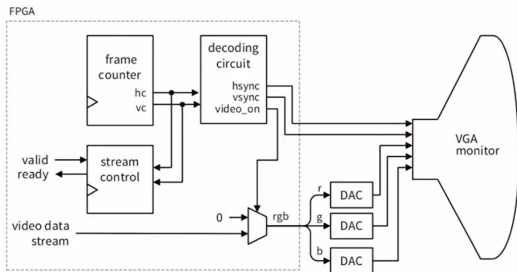
$$\text{pixel clock rate} = p \times l \times s$$

- p : Number of **p**ixels in a horizontal scan line
- l : Number of **l**ines in a screen
- s : Number of **s**creens per second

Compute the clock rate for the
VGA resolution (640x480)



VGA (Video Graphics Array)



Artix-7



VGA (Video Graphics Array)

8	4	2	1	8	4	2	1
.	.	.	#	#	.	.	.
.	.	#	#	#	#	.	.
.	#	#	#	#	#	.	.
#	#	.	#	#	.	#	#
#	#	#	#	#	#	#	#
.	.	#	.	.	#	.	.
.	#	.	#	#	.	#	.
#	.	#	.	.	#	.	#

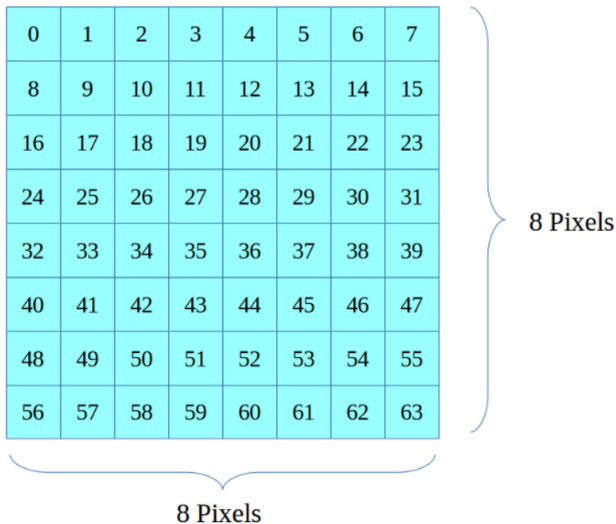
0x18
0x3C
0x7E
0xDB
0xFF
0x24
0x5A
0xA5

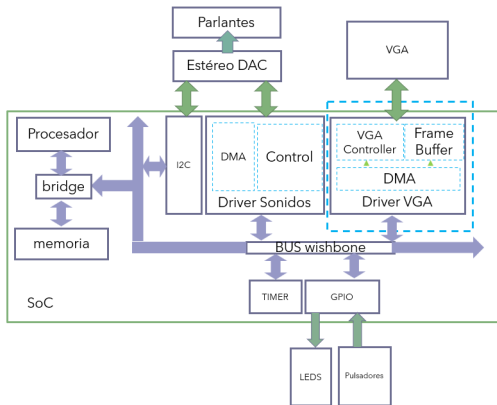
8	4	2	1	8	4	2	1
.	#	.	.	.	.	#	.
.	.	#	.	.	#	.	.
.	#	#	#	#	#	.	.
.	#	.	#	#	.	#	.
#	#	#	#	#	#	#	#
#	#	#	#	#	#	#	#
.	.	#	.	.	#	.	.
.	#	.	.	.	.	#	.

0x42
0x24
0x7E
0x5A
0xFF
0xFF
0x24
0x42



VGA (Video Graphics Array)





SoC Space Invaders

Mapa de Memoria General

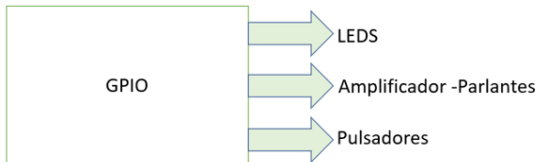
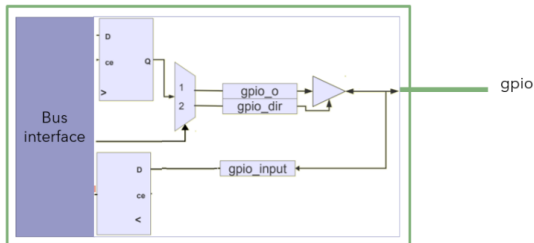
Mapa de memoria general
VGA
I2C
TIMER
GPIO
MAIN_RAM
SRAM
ROM



Mapa de Memoria - GPIO

Mapa de memoria del GPIO

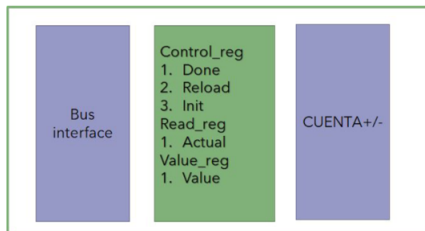
WRITE	Base+08	W
READ	Base+04	R
DIR	Base+00	W



Mapa de Memoria - TIMER

Mapa de memoria del TIMER

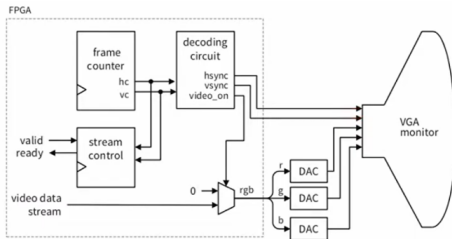
CTRL	Base+08	R
COUNT	Base+04	R
LIMIT	Base+00	W



Mapa de Memoria - VGA

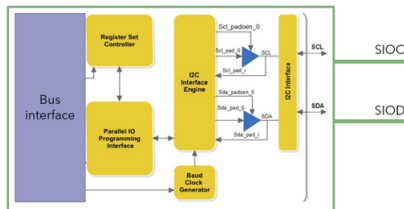
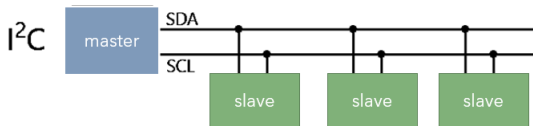
Mapa de memoria VGA

RGB	Base+0F00	W
DIR	Base+0F0C	W



Mapa de Memoria - I2C

Mapa de memoria I2S		
CTR	Base+08	W/R
SDA	Base+04	W/R
SCL	Base+00	W/R



References

-  H. Kopka and P. W. Daly, *Referencia 1*, 3rd ed. Harlow, England: Addison-Wesley, 1999.
-  H. Kopka and P. W. Daly, *Referencia 2*, 3rd ed. Harlow, England: Addison-Wesley, 1999.
-  H. Kopka and P. W. Daly, *Referencia 3*, 3rd ed. Harlow, England: Addison-Wesley, 1999.
-  "Space Invaders"" . Simple CPU design.
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-  nas Salah Eddin. M14 - 1 - Introduction to Video Displays. (17 de noviembre de 2020). Accedido el 11 de abril de 2023. [Video en línea]. Disponible: <https://www.youtube.com/watch?v=Qlybt6PHK8s>
-  nas Salah Eddin. M14 - 2 - Stream Interface. (17 de noviembre de 2020). Accedido el 11 de abril de 2023. [Video en línea]. Disponible: <https://www.youtube.com/watch?v=Qlybt6PHK8s>